

Join the lecture online on *your* dashboard.

Let's start with a minute of silence.

आचार्यात् पादं आधत्ते पादं शिष्यः स्वमेधया । पादं सब्रह्मचारिभ्यः पादं कालक्रमेण च ॥

Meaning:

A student acquires knowledge in four equal parts:

- One-fourth from the teacher
- One-fourth through self-reflection and independent thinking
- One-fourth through discussions with peers and fellow learners
- One-fourth over time through personal experience



Projects for this Unit





Create



Home



Competitions



Datasets



Models



Benchmarks



Game Arena



Code



Discussions



Learn



More



Your Work



VIEWED



Hedge fund - Time series forecasting



Forecasting and Anomaly Detection



Hourly Energy Consumption









A DATA COMPANY - COMMUNITY PREDICTION COMPETITION - 3 MONTHS TO GO

Join Competition

...

Hedge fund - Time series forecasting

Time series forecasting by code, sub-code, sub_category and horizon

Overview

Data

Code

Models

Discussion

Leaderboard

Rules

Overview

Participants will receive an integer-indexed time series dataset (ts_index column), where each record is identified by a code, sub-code, sub-category and forecast horizon. Your objective is to train a model that generalizes robustly out-of-sample and accurately predicts future values for each combination (code, sub-code, sub-category, horizon). Your forecast will not use any data whose ts_index is greater than the ts_index of the forecast data. Submissions are ranked according to an aggregate out-of-sample metric calculated for all combinations.

Start

6 days ago

Close

3 months to go

Competition Host

A data COMPANY

Prizes & Awards

\$10,000

Does not award Points or Medals

Participation

283 Entrants

43 Participants

43 Teams

76 Submissions

Tags

Custom Metric

Projects for this Unit





Create



Home



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Models



Benchmarks



Game Arena



Code



Discussions



Learn



More



Your Work



VIEWED



Store Sales - Time S...



Search

Store Sales - Time Series Forecasting

Use machine learning to predict grocery sales

Overview

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Code

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Overview



This competition runs indefinitely with a rolling leaderboard. [Learn more](#)

Description





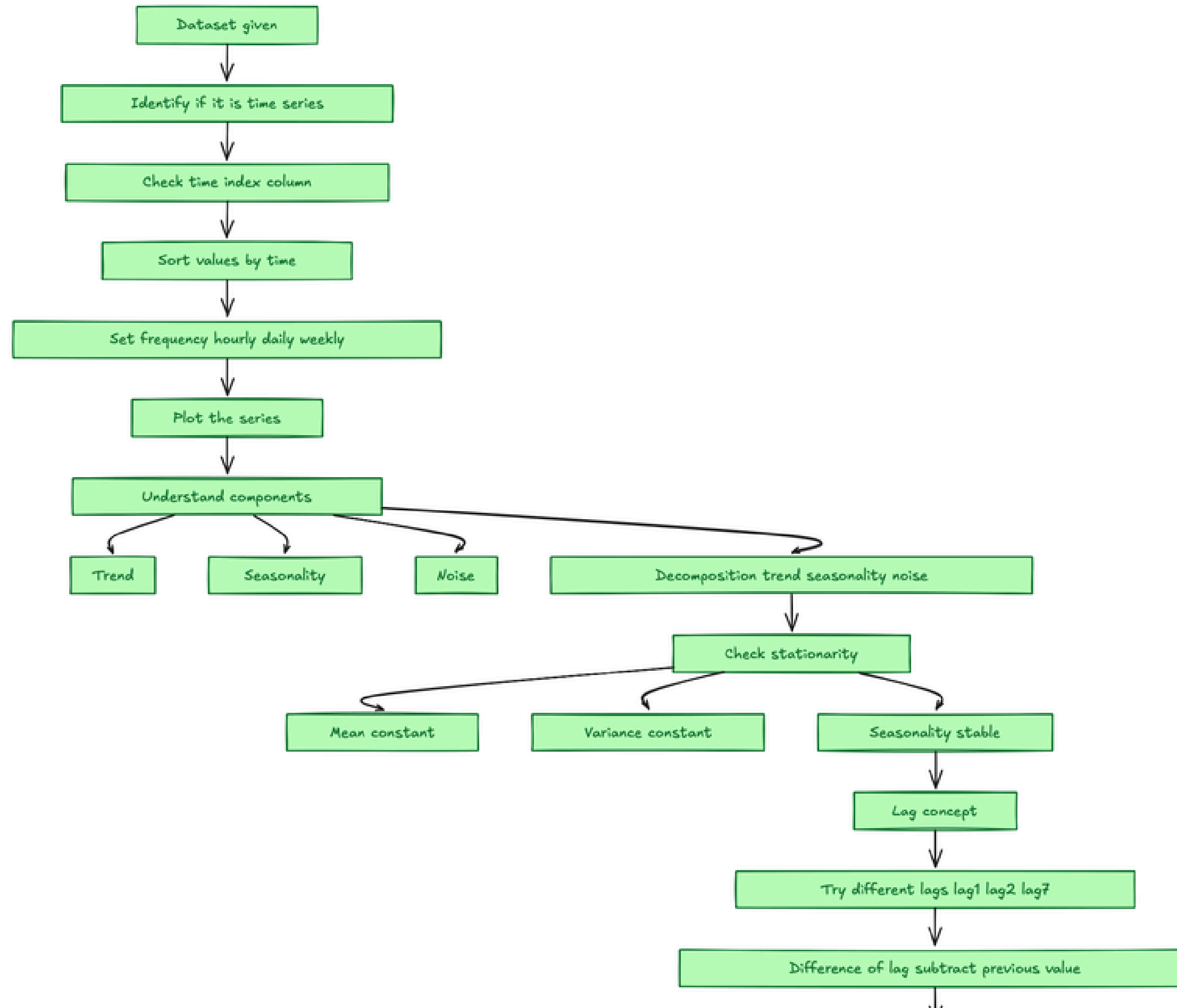
Goal of the Competition

In this "getting started" competition, you'll use time-series forecasting to forecast store sales on data from Corporación Favorita, a large Ecuadorian-based grocery retailer.

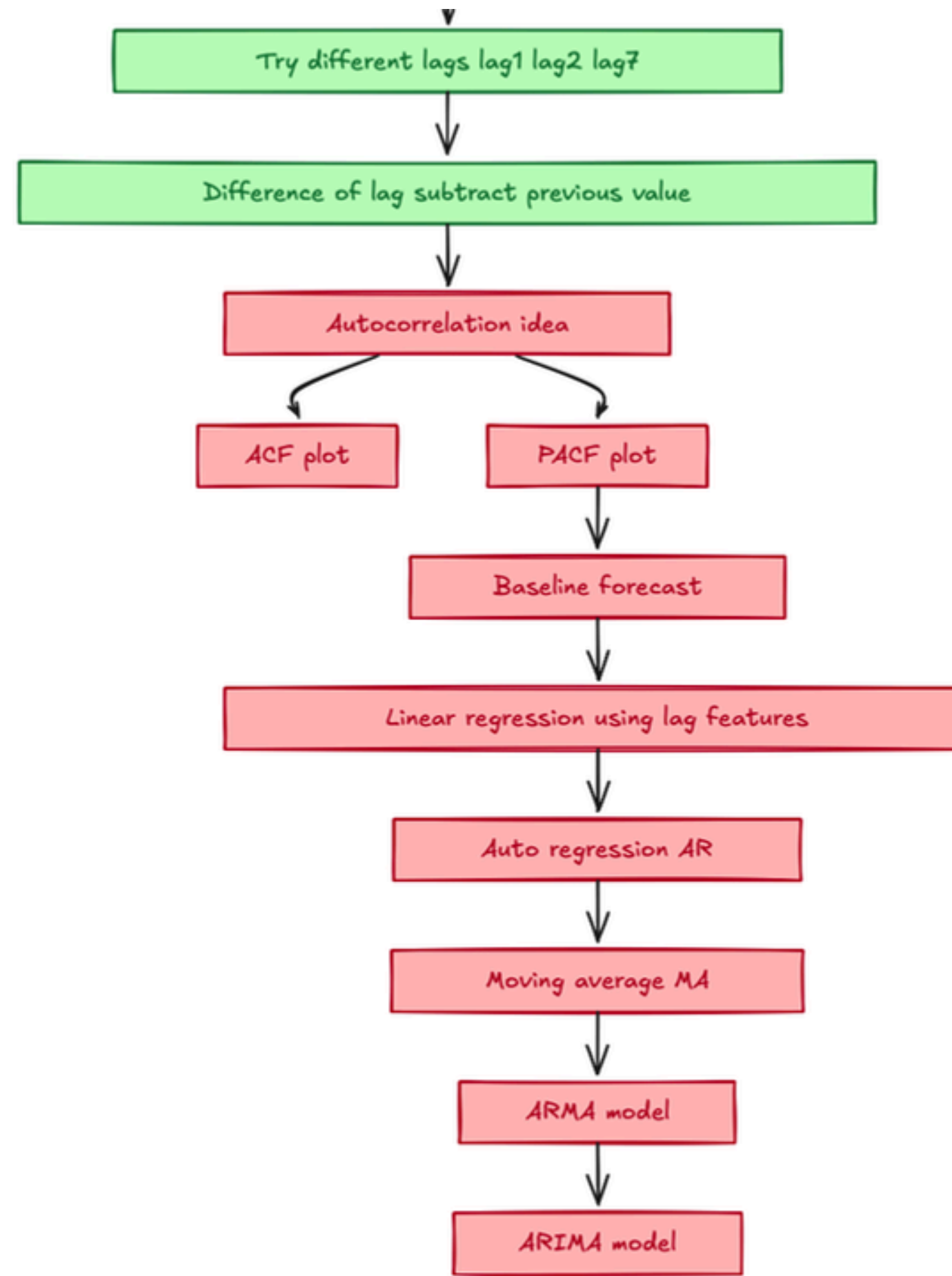
Specifically, you'll build a model that more accurately predicts the unit sales for thousands of items sold at different Favorita stores. You'll practice your machine learning skills with an approachable training dataset of dates, store, and item information, promotions, and unit sales.

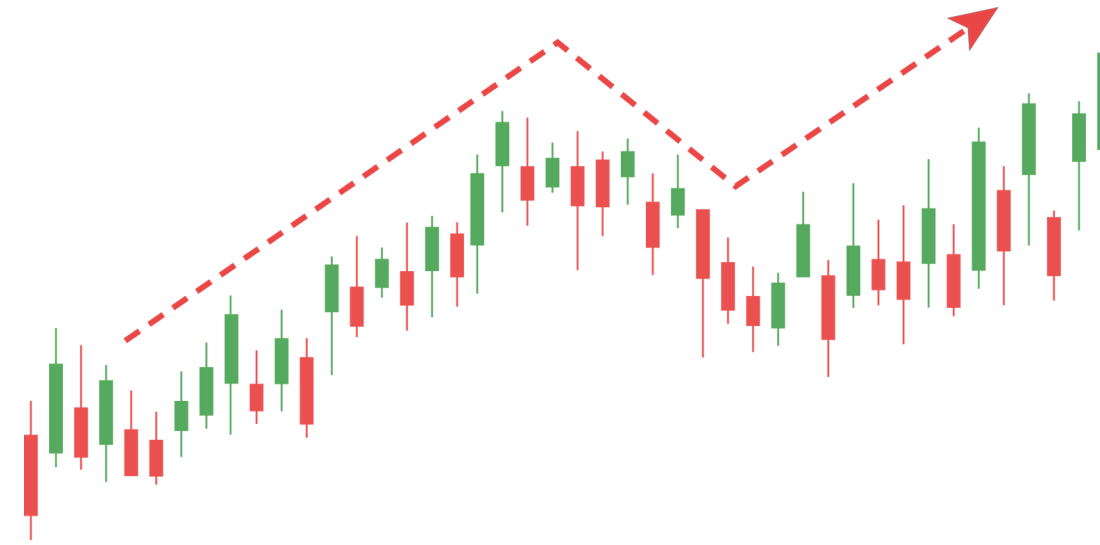
149

Recap

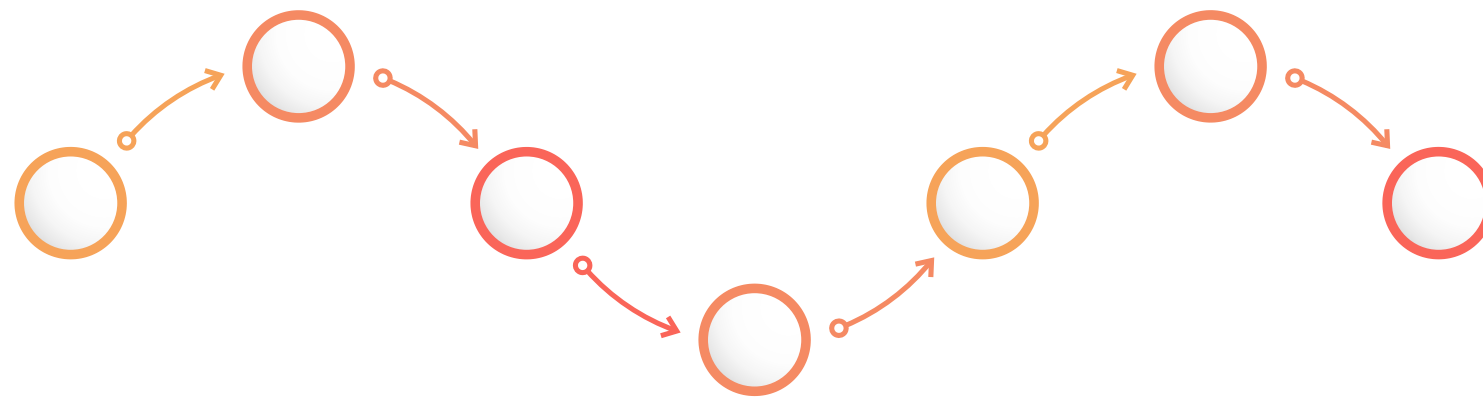


Agenda



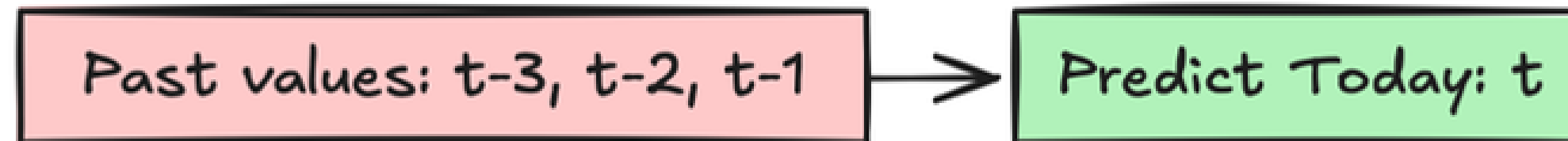


In the previous lecture, we learned how **present** value **depends** on the **past** data and why the **order of time** is **important**. What strategy will you use to predict the present values?



Question

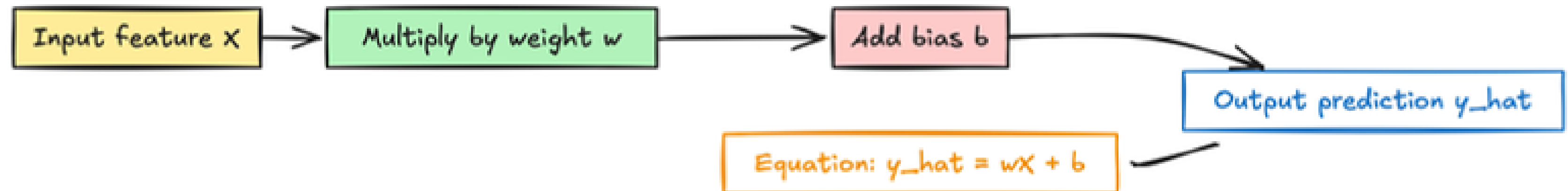
Can you think of a method to predict the value of **today's data point** using **previous data points**.



Question: Can you predict Sales at 6th time stamp?

Day	Sales
1	120
2	135
3	128
4	150
5	160
6	???

Can you relate it with linear regression?



$$Y = wX + b$$

Question

Day	Sales
1	120
2	135
3	128
4	150
5	160
6	???

What is X and Y in this?

Question

Day	Sales
1	120
2	135
3	128
4	150
5	160
6	???

Can we consider Day as X?

Question

Day	Sales
1	120
2	135
3	128
4	150
5	160
6	???



**Can we consider Values of Sales from Day 1 to Day 6
as X?**

Question

Day	Sales	Sales (Lag 1)
1	120	
2	135	120
3	128	135
4	150	128
5	160	150
6	???	

Lag 1 – Can we consider Values of Sales from Day 1 to Day 6 as X?

SOLUTION: Linear Regression using Lags

$$y_t = c + \phi y_{t-1} + \epsilon$$

Diagram illustrating the components of the linear regression equation with lags:

- y_t : Current Target Value
- c : A constant
- ϕ : Influencing Factor
- y_{t-1} : Value of target 1 day ago
- ϵ : White noise

We can use the lags and develop a relation of the target variable with the specific past values (lags).

Can we **incorporate** the **importance** of
all the **other lags**?

Question

Day	Sales	Sales (Lag 1)	Sales (Lag 2)
1	120		
2	135	120	
3	128	135	120
4	150	128	135
5	160	150	128
6	???		

Can Lag 1 and Lag 2 Act as X_1 and X_2 ?

Equation for prediction (other lags)

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t$$

y_t : (Value at time t)

c : (Constant/Intercept)

ϕ_i : (Coefficient for lag i)

ϵ_t : (White noise/Error term)

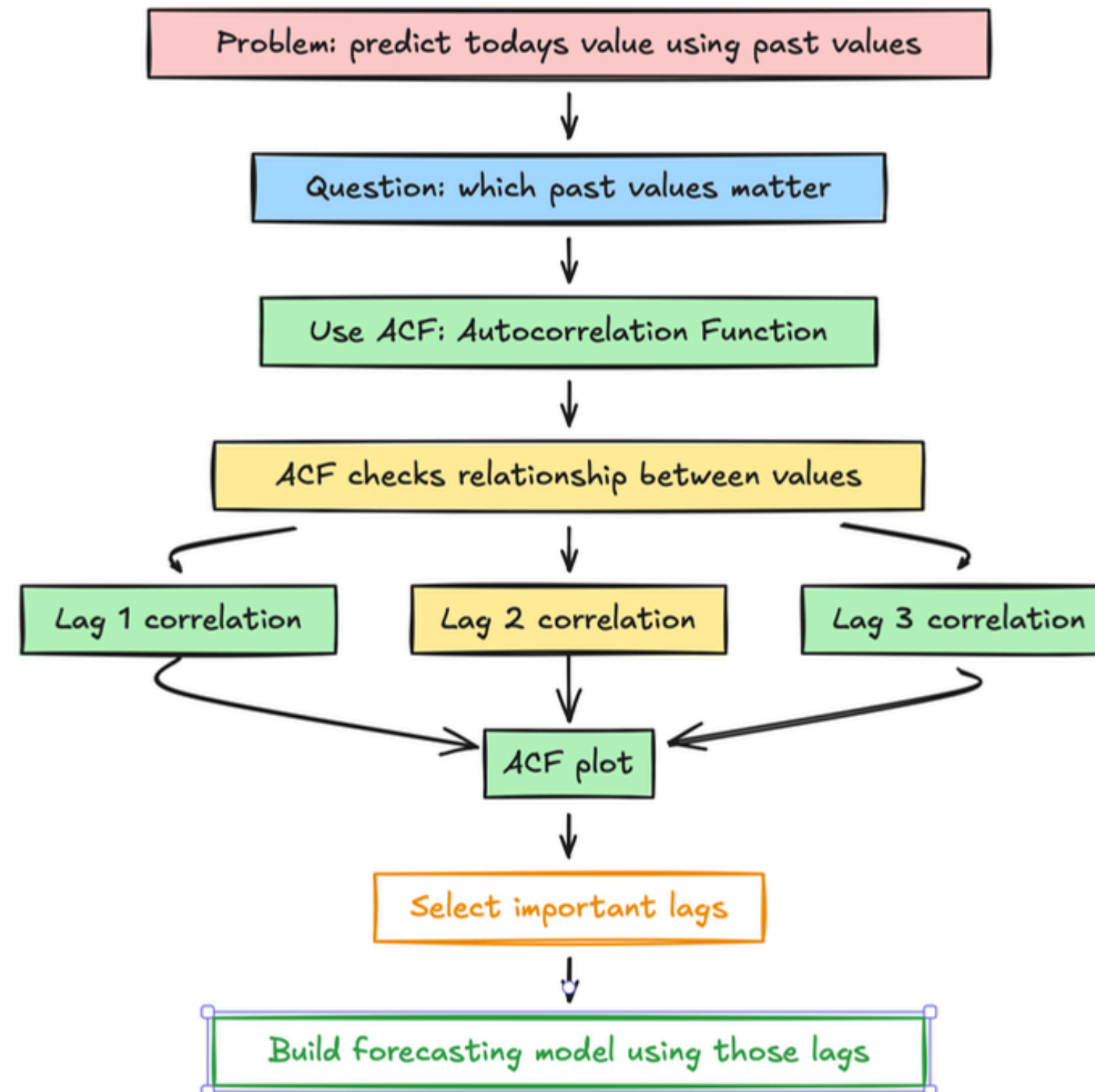
Equation for prediction (other lags)

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t$$

- How can we **determine the values of lag coefficients** and decide which **lags** to include in our model?
- Including **all lags** could lead to unnecessary complexity.
- Therefore, it's **crucial** to **select** only the most **relevant lags** that **significantly influence** the time series, ensuring the model remains both efficient and interpretable.

How to solve this problem?

We use Autocorrelation Function to solve this problem.



Autocorrelation Function (ACF)

- The ACF is a **collection of autocorrelations** across **many lags**, shown together.
- ACF answers:
 - How strong is **lag 1**?
 - How strong is **lag 2**?
 - How strong is **lag 3**?
 - How strong is **lag k**?
- The **ACF** is usually displayed as a **plot of autocorrelations** at **different lags**.
- Think of it like this:
 - **Autocorrelation** = one value
 - **ACF** = list of autocorrelations for multiple lags

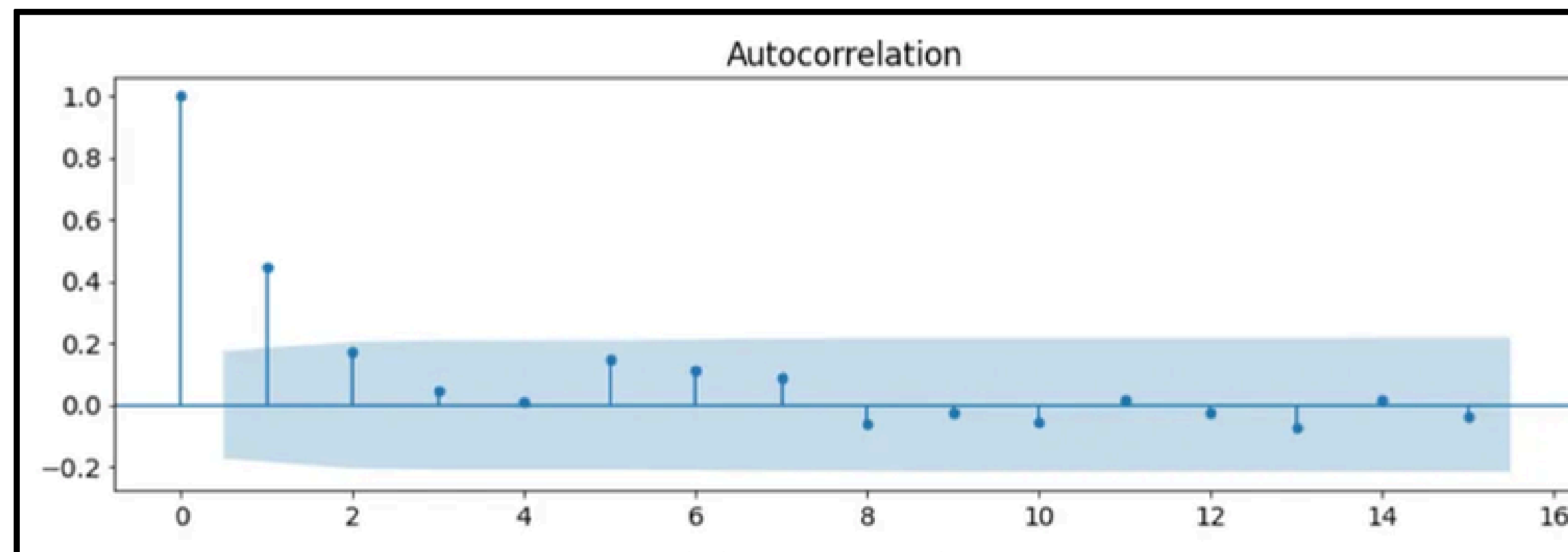
Range of ACF

Because ACF is literally a **correlation coefficient**, and correlation is always bounded:

$$-1 \leq \rho(k) \leq 1$$

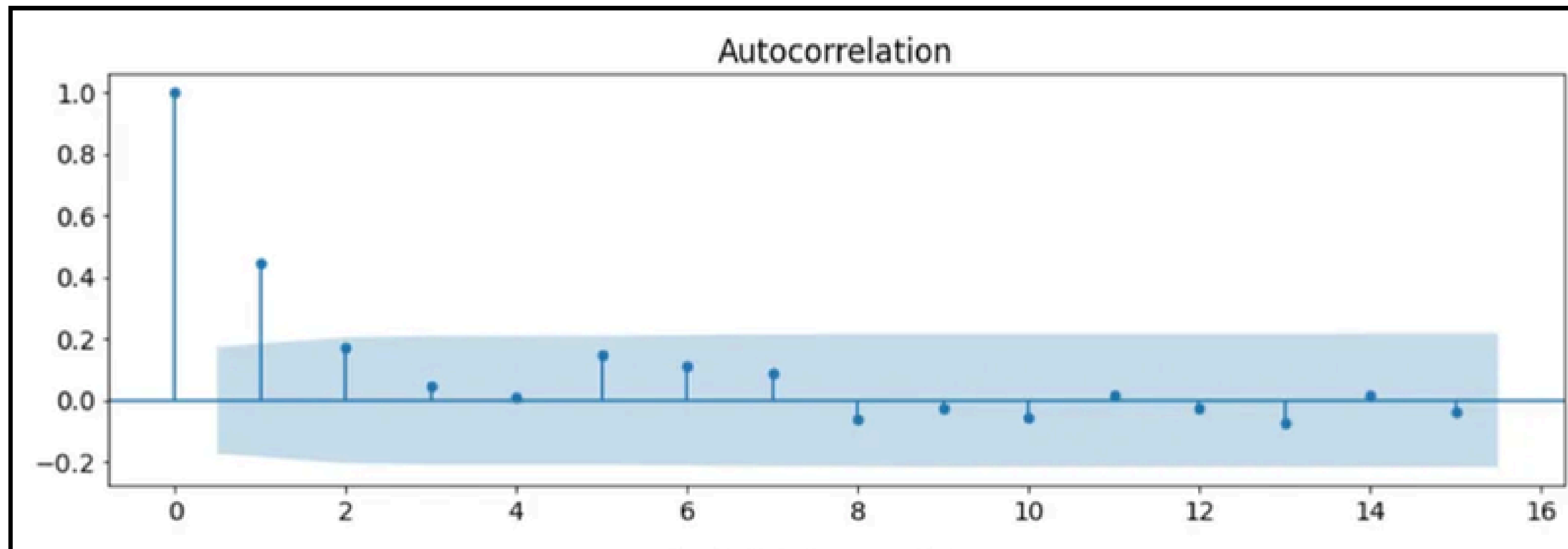
Interpretation:

- $\rho(k) \approx 1$: strong positive relation (moves together)
- $\rho(k) \approx -1$: strong negative relation (moves opposite)
- $\rho(k) \approx 0$: no relation



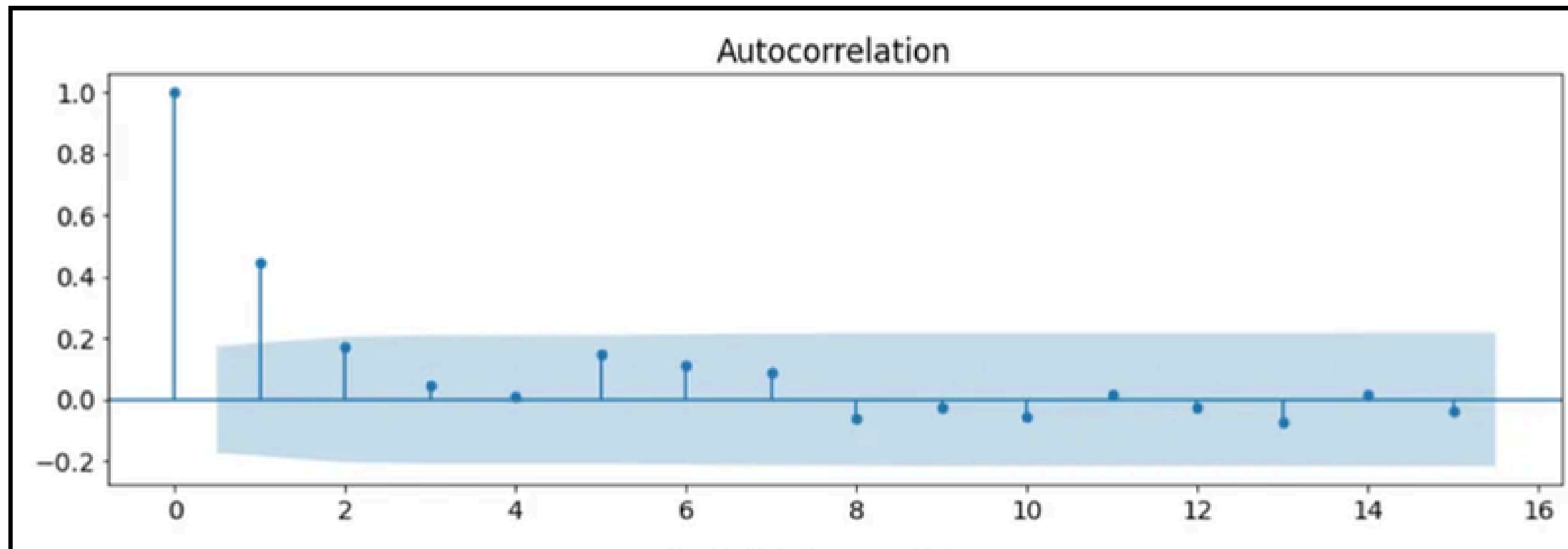
Autocorrelation Function Plot

- Each vertical spike represents the strength of correlation at that lag.
- The shaded **blue** region represents the **confidence band**:
 - Bars inside the box → NOT statistically significant
 - Bars outside the band → statistically significant

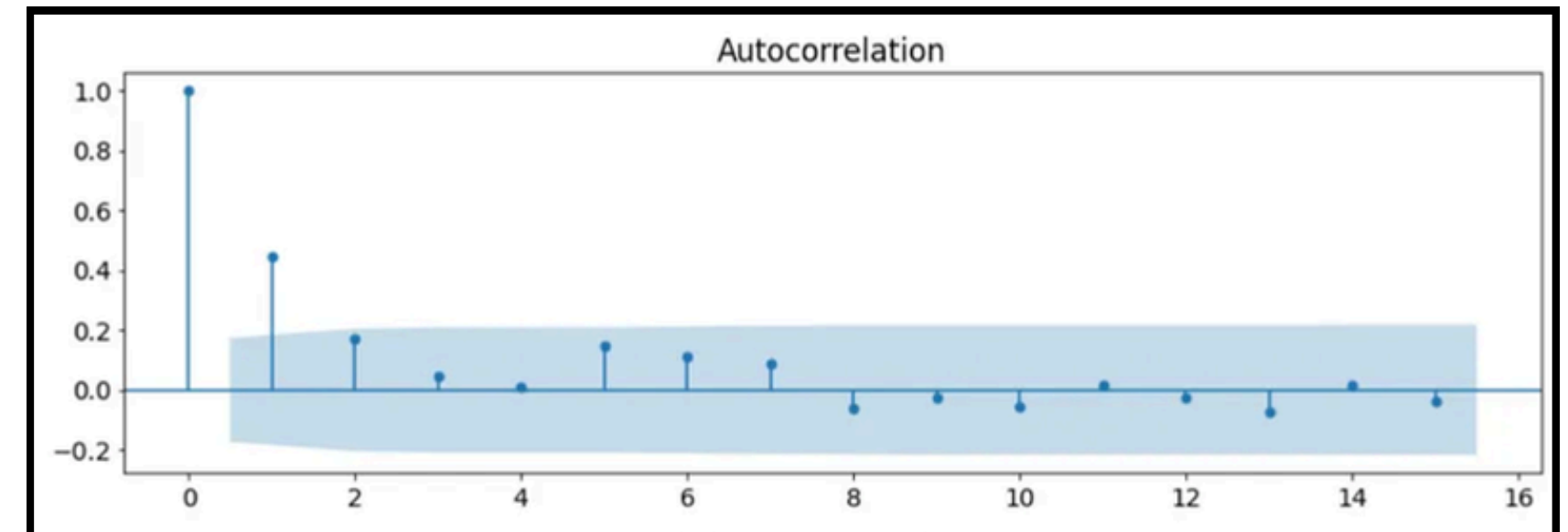
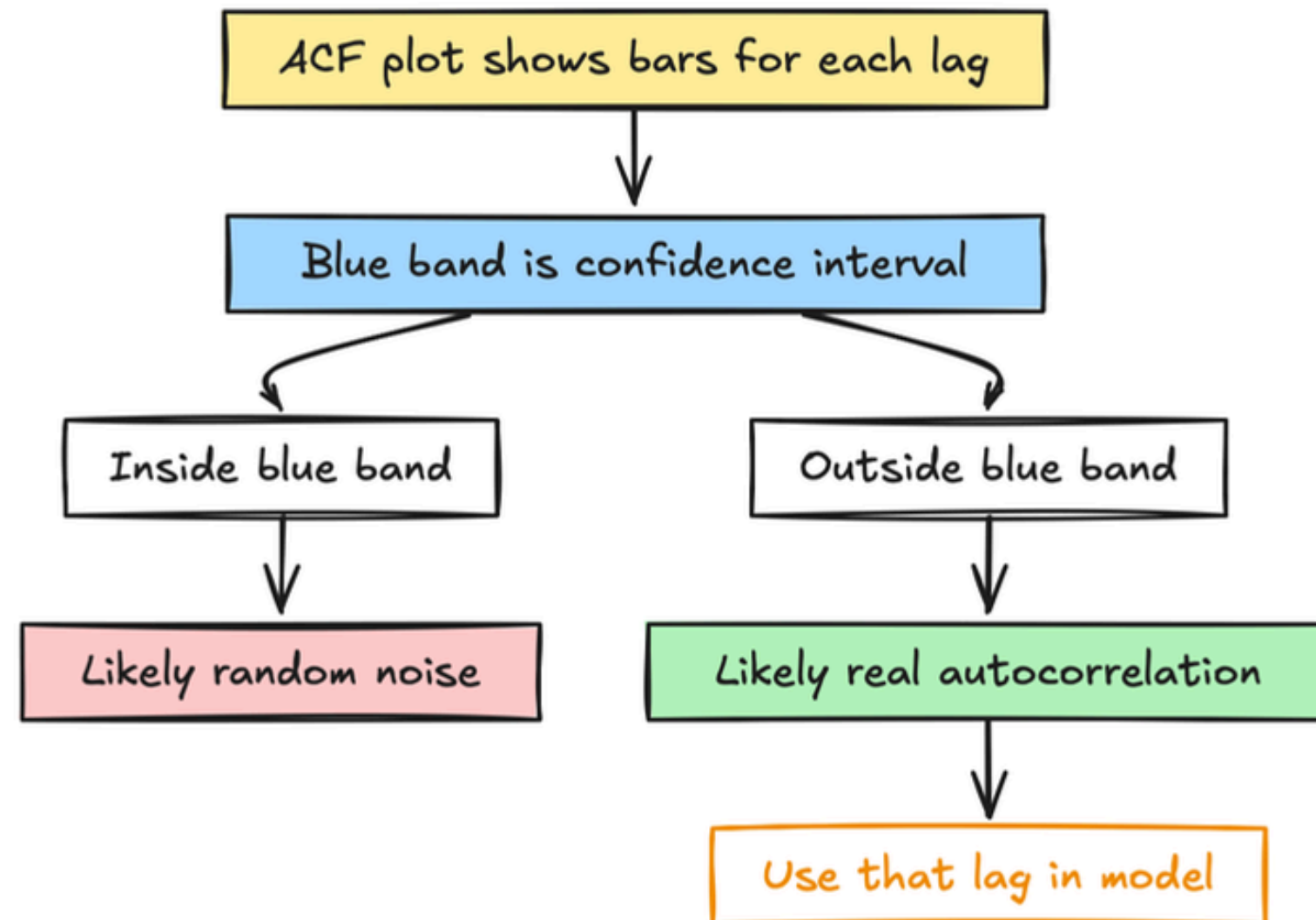


What is confidence band?

- So the plot shows a blue shaded region = range of correlations that are “normal by chance”.
- If a spike goes outside the blue band, it is likely a real signal (statistically significant autocorrelation).
- Most ACF plots use an approximate 95% confidence interval

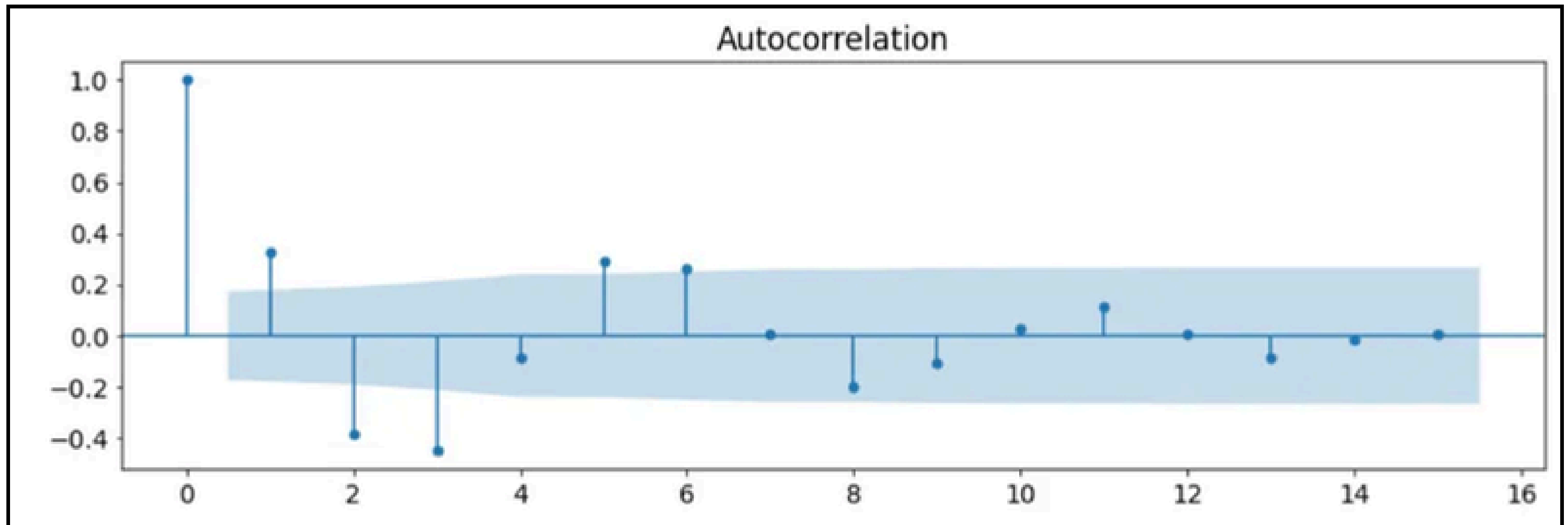


What is confidence band?



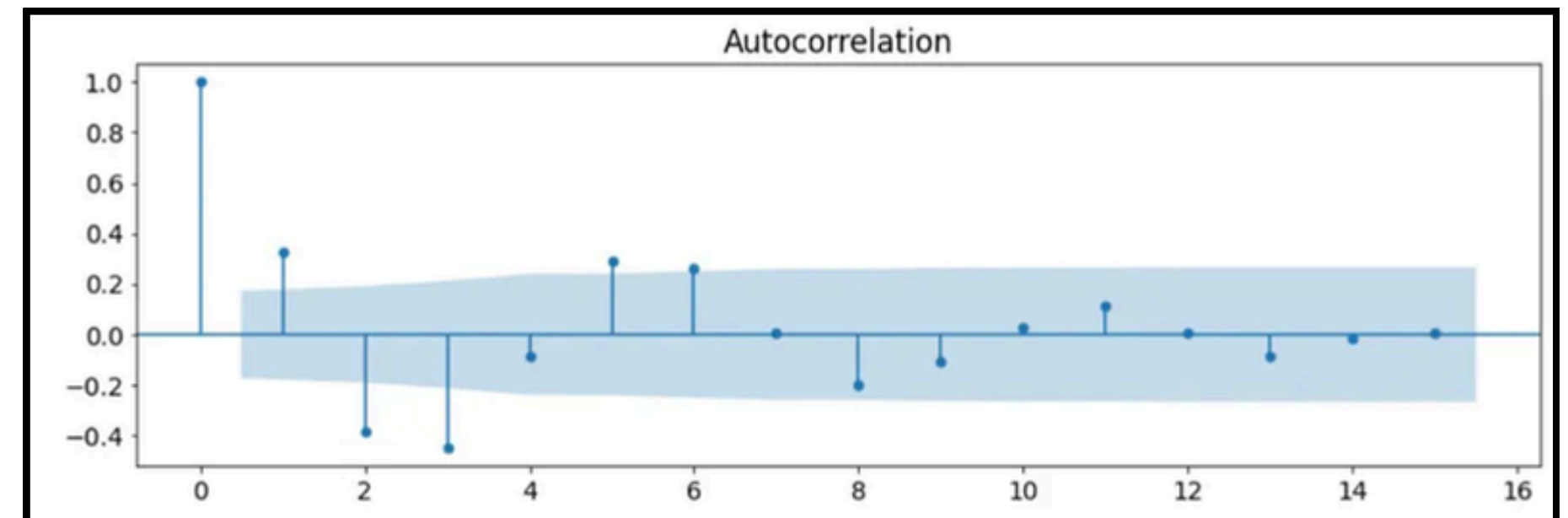
Problem Statement

What does the negative bar in the following plots mean ?



Autocorrelation Function Plot

- A **negative autocorrelation** at **lag 2** means:
- The value **today tends to move** in the **opposite direction** from the value from two time steps ago.
- In simpler terms:
 - If the value 2 days ago was high,
 - **Today tends to be lower.**
 - If the value 2 days ago was low,
 - **Today tends to be higher.**
- This indicates a **reversal effect** or **correction pattern** after two time steps.



Autocorrelation Function Plot

Most ACF plots use an approximate 95% confidence interval:

$$CI \approx \pm \frac{1.96}{\sqrt{N}}$$

Where:

- N = number of observations in the time series
- 1.96 = z value for 95% confidence

So:

$$\text{Blue band upper} = + \frac{1.96}{\sqrt{N}}$$

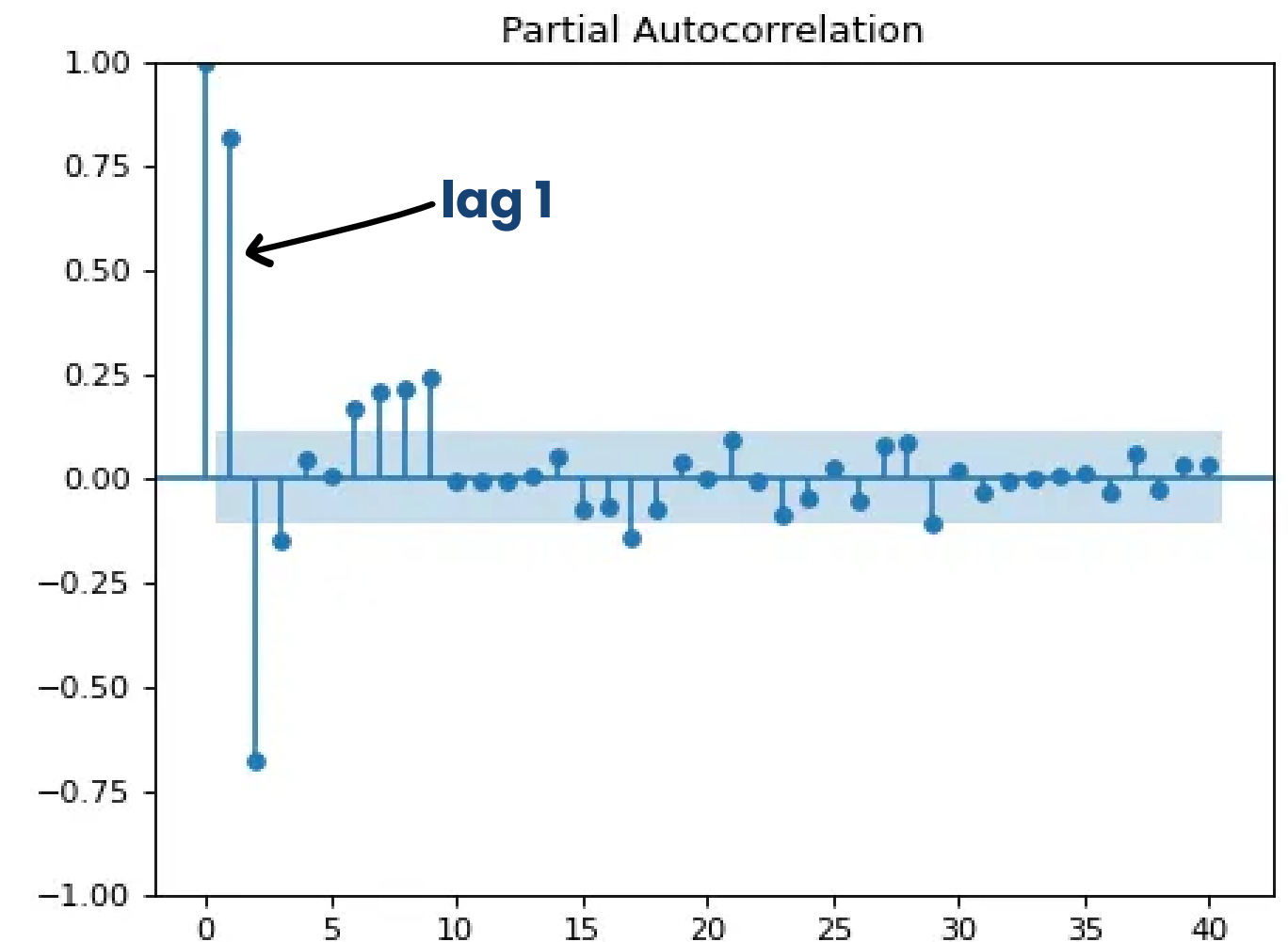
$$\text{Blue band lower} = - \frac{1.96}{\sqrt{N}}$$

SOLUTION: Linear Regression using Lags

$$y_t = c + \phi y_{t-1} + \epsilon$$

Diagram illustrating the components of the linear regression model using lags:

- y_t : Current Target Value
- c : A constant
- ϕ : Influencing Factor
- y_{t-1} : Value of target 1 day ago
- ϵ : White noise



• ~~$\lg 1 = 0.8$~~

$$\lg 2 = 0.75$$

$$\lg 3 = 0.9$$

$$C_i = 0.79$$

Partial Autocorrelation (PACF)

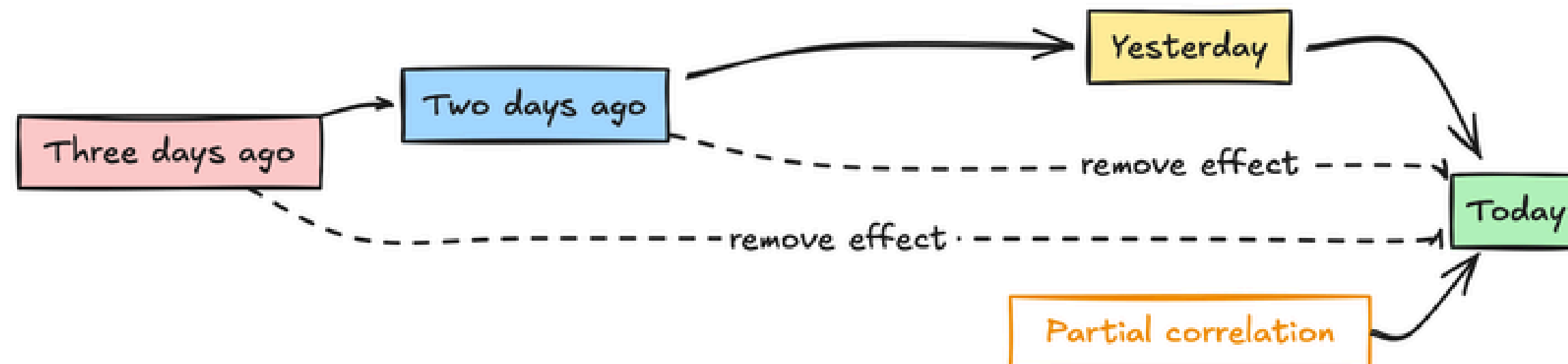
- Imagine you're predicting today's Zomato orders:
 - **Yesterday's orders** influence **today** → **lag 1**
 - The day before yesterday also influences today,
 - But part of that influence is already explained by lag 1.
- PACF removes all “**shortcut**” effects.
- **ACF says:**
 - “How related are Day 3 and Day 1 overall?”
- **PACF says:**
 - “How related are Day 3 and Day 1 if I ignore what happened on Day 2?”

ACF Vs PACF

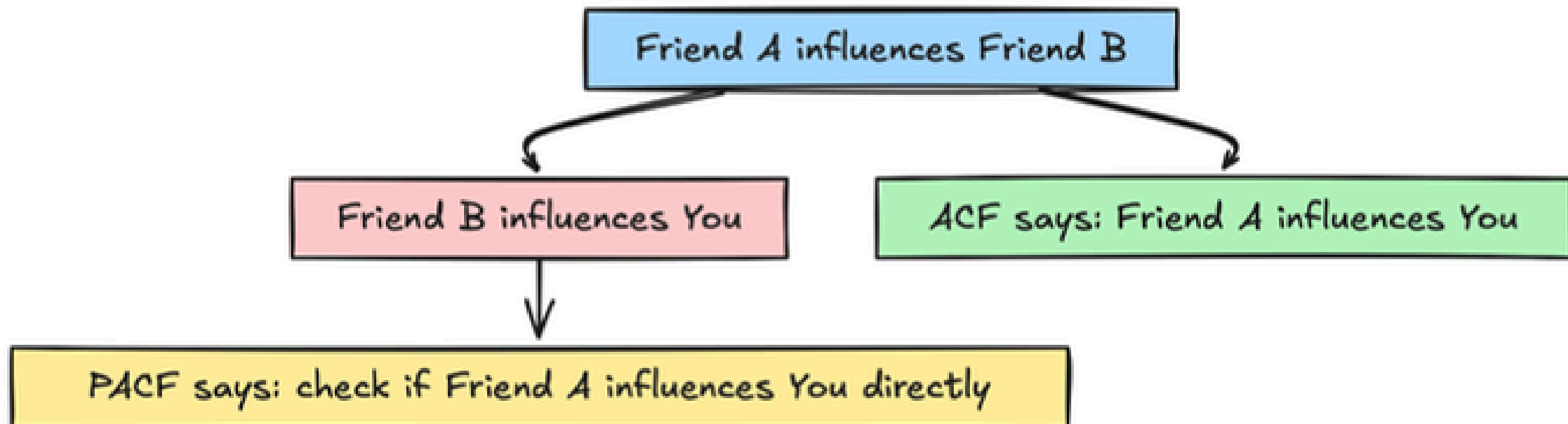
**ACF shows
total influence**



**PACF shows
direct influence**

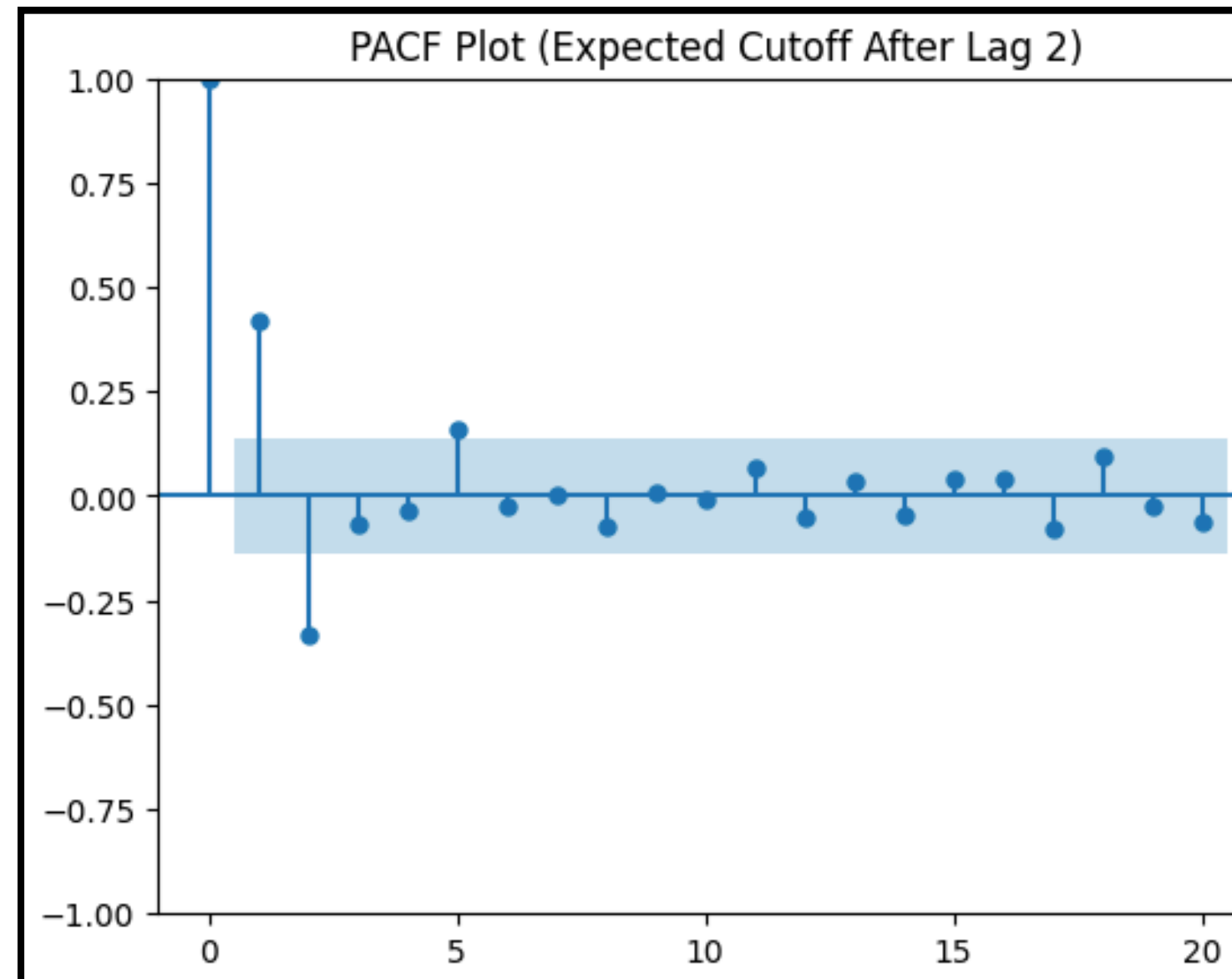


ACF Vs PACF



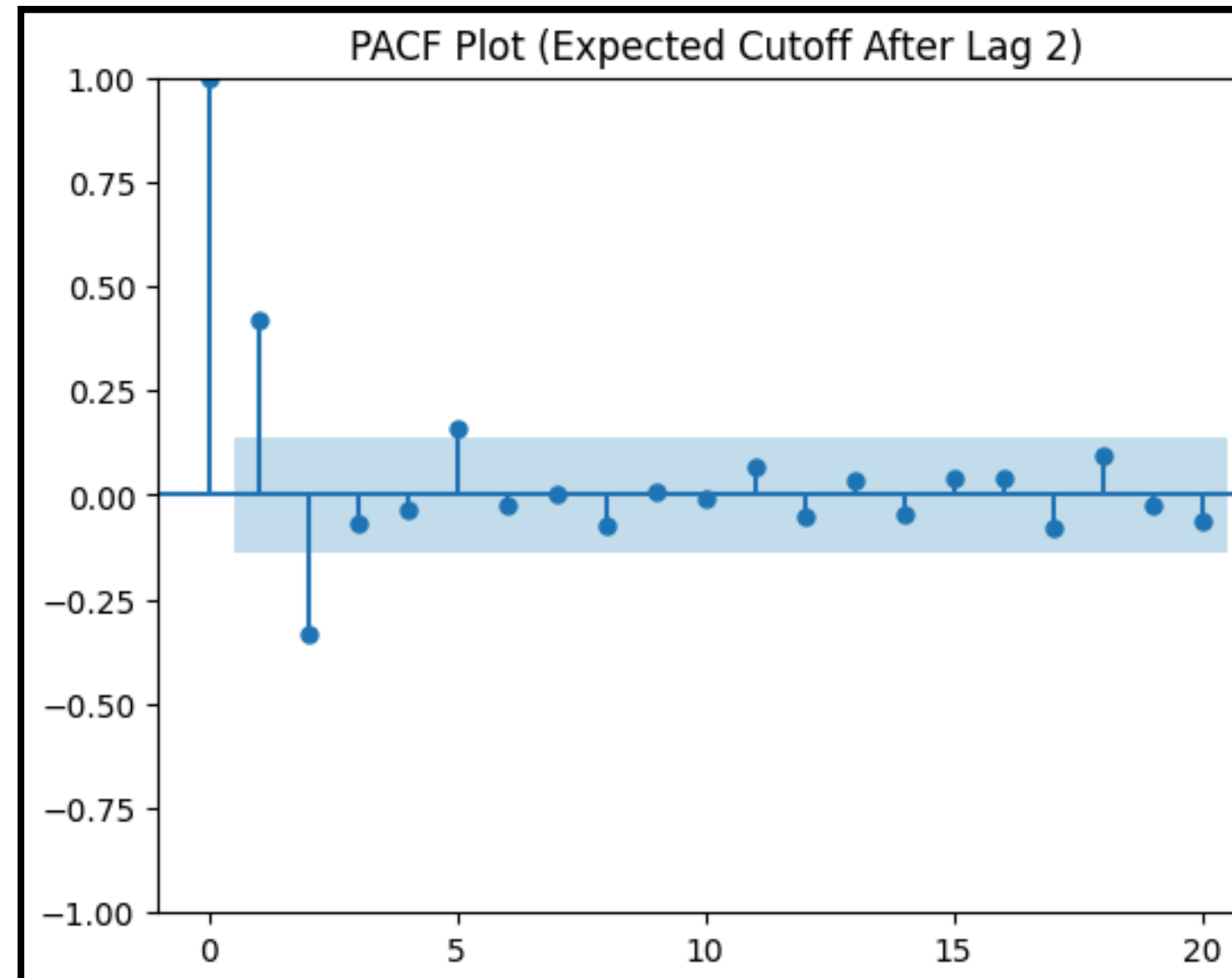
Interpretation of PACF Plot

You are analyzing a **daily sales time series** that has already been made **stationary** (trend and seasonality removed). You compute the Partial Autocorrelation Function (PACF) and obtain the following plot:



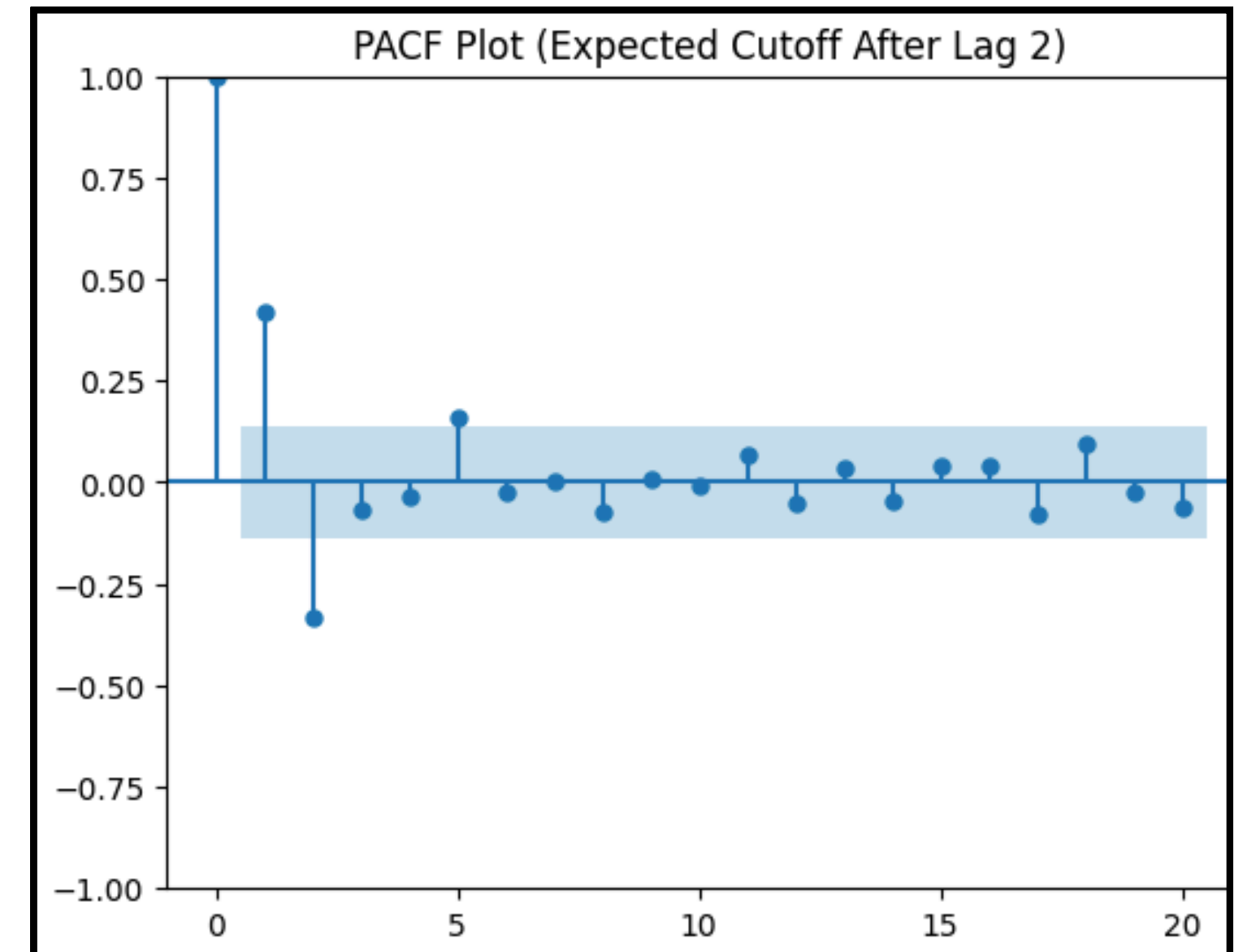
Interpretation of PACF Plot

What does this **PACF plot suggest** about **how current sales** depend on **past sales**?

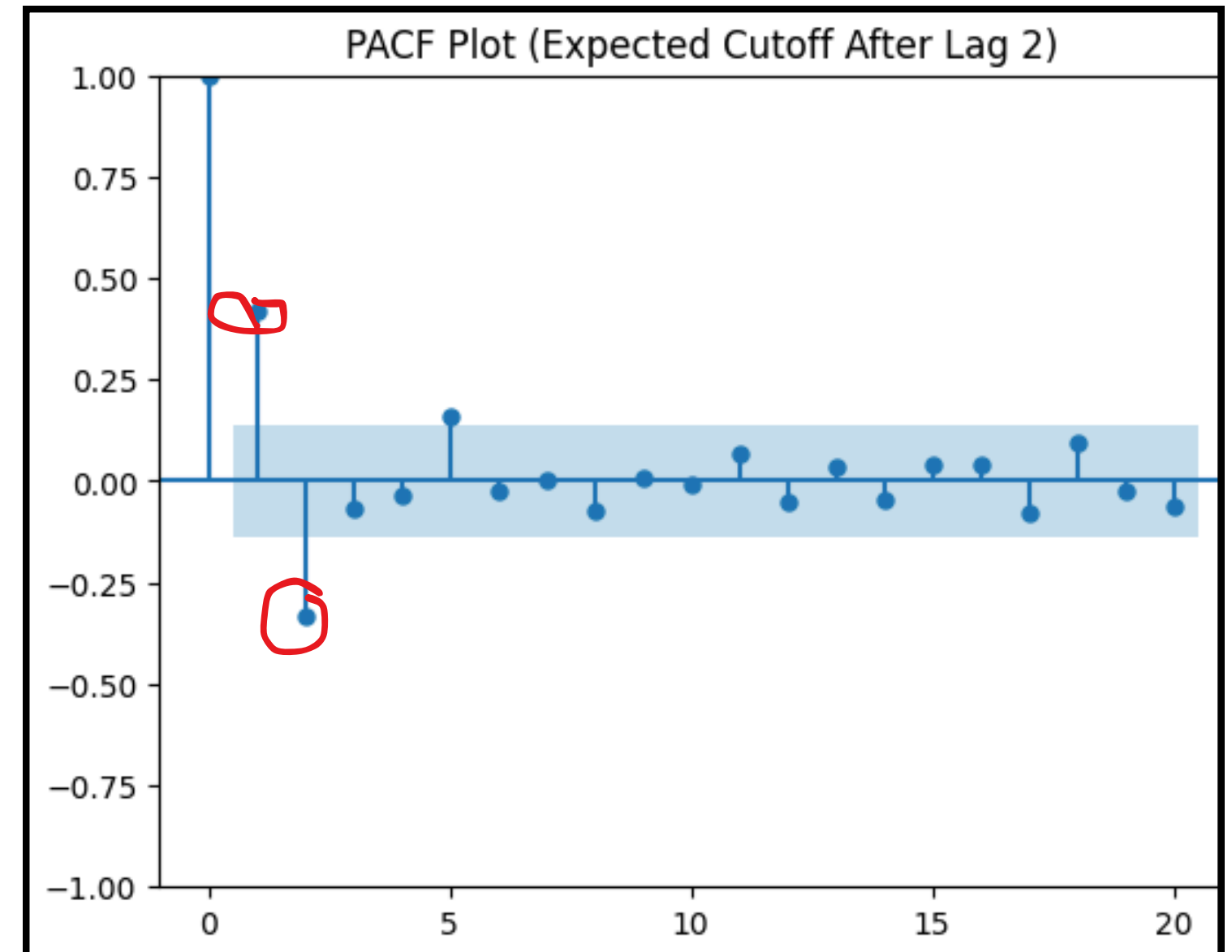
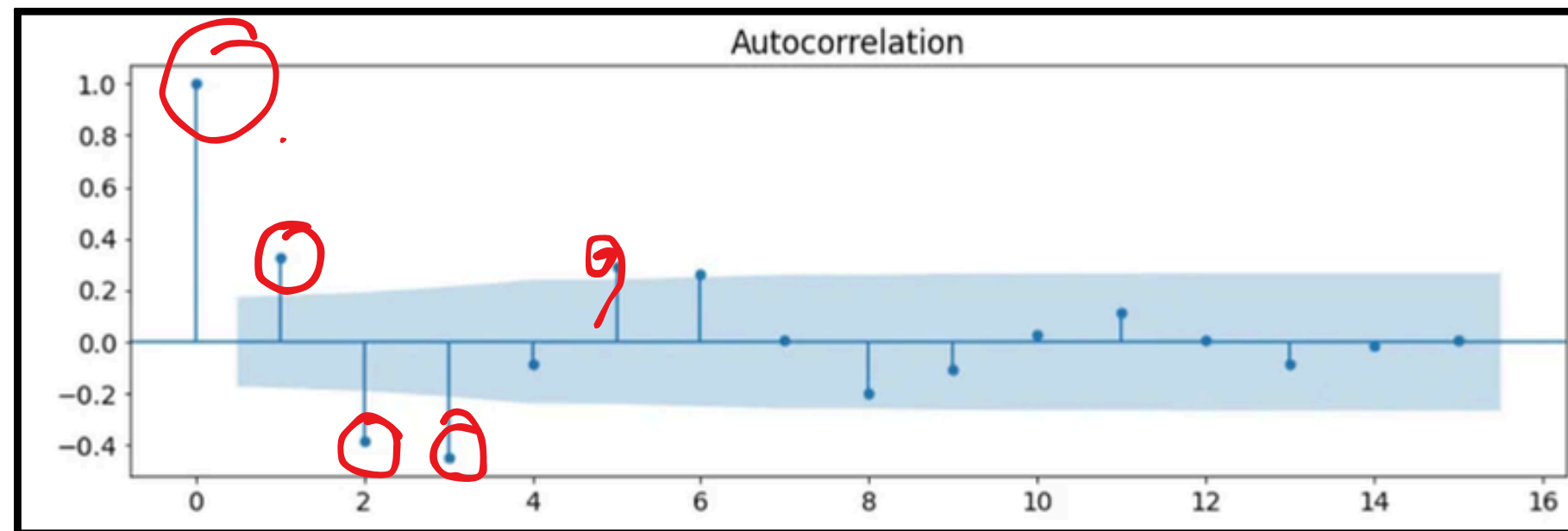


Interpretation of PACF Plot

- **Lag 1:** Extreme positive spike
 - Yesterday has a strong direct influence on today.
- **Lag 2:** Strong negative spike
 - A negative PACF at lag 2 tells you:
 - Day-before-yesterday has a strong inverse effect on today, once we remove yesterday's influence.
- **Lags 3 afterwards:** All small, inside the confidence band
 - Once you go past lag 2, no more significant spikes
 - Everything is basically noise
 - No long-range "direct" relationships



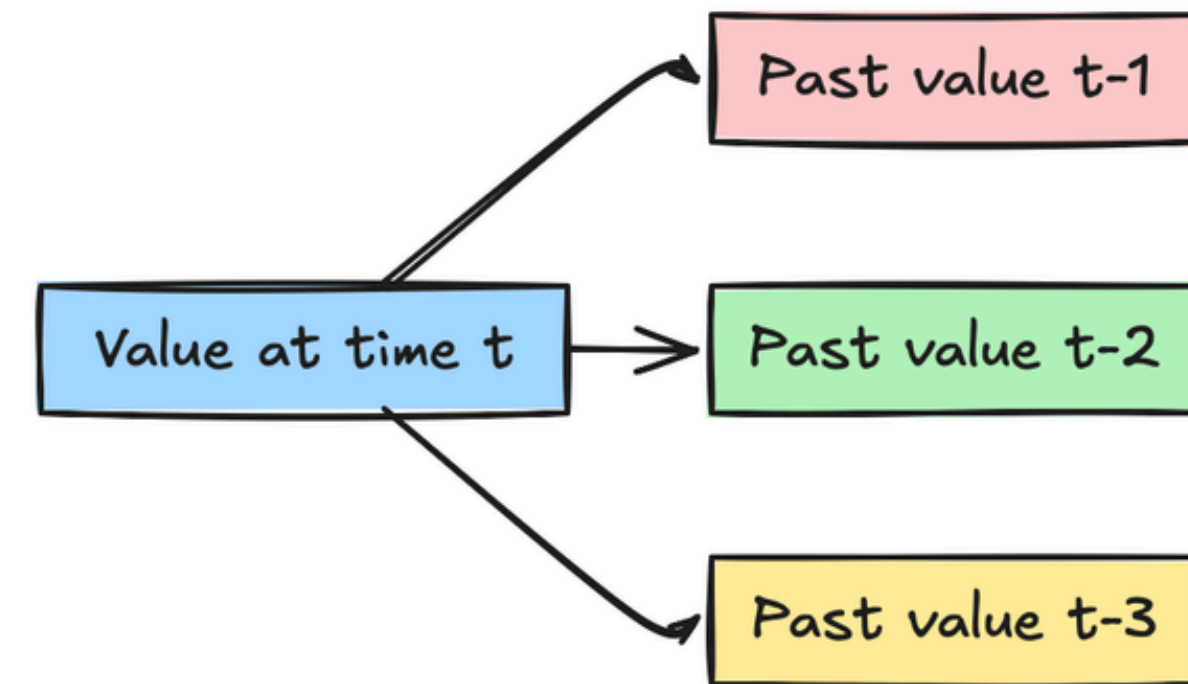
Interpretation of PACF Plot



What We just **discovered** is
Autoregression

Auto Regression

- **Auto** (Self): implies the variable is **predicted** by itself (specifically, its own past values).
- **Regression**: a **statistical technique** for predicting a **target value** from a **combination of inputs**.
- Unlike **standard regression** (which uses other features like x_1 , x_2 to predict y), Autoregression **uses past values** as the features.

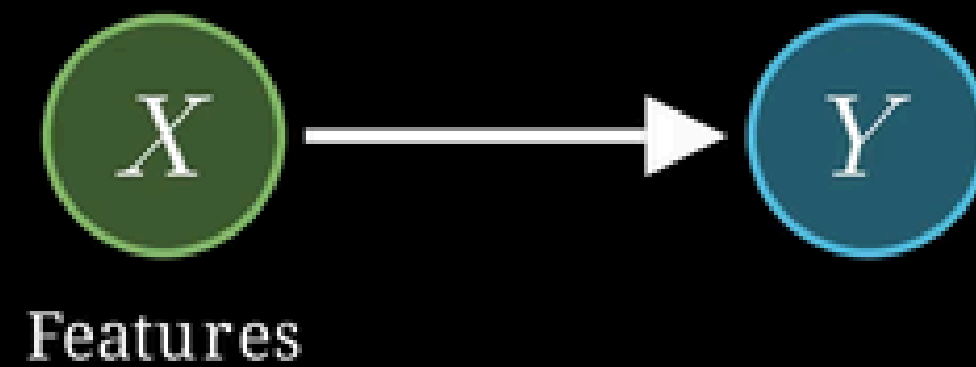


Regression Vs AutoRegression

Regression

Linear Prediction

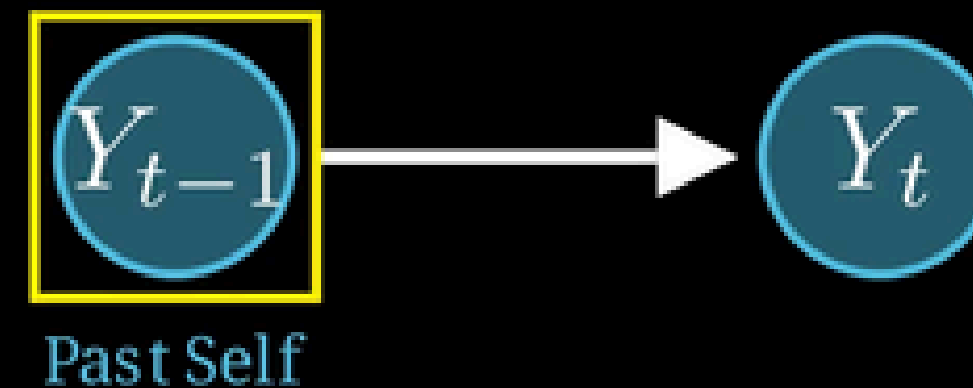
Standard Regression



AutoRegression

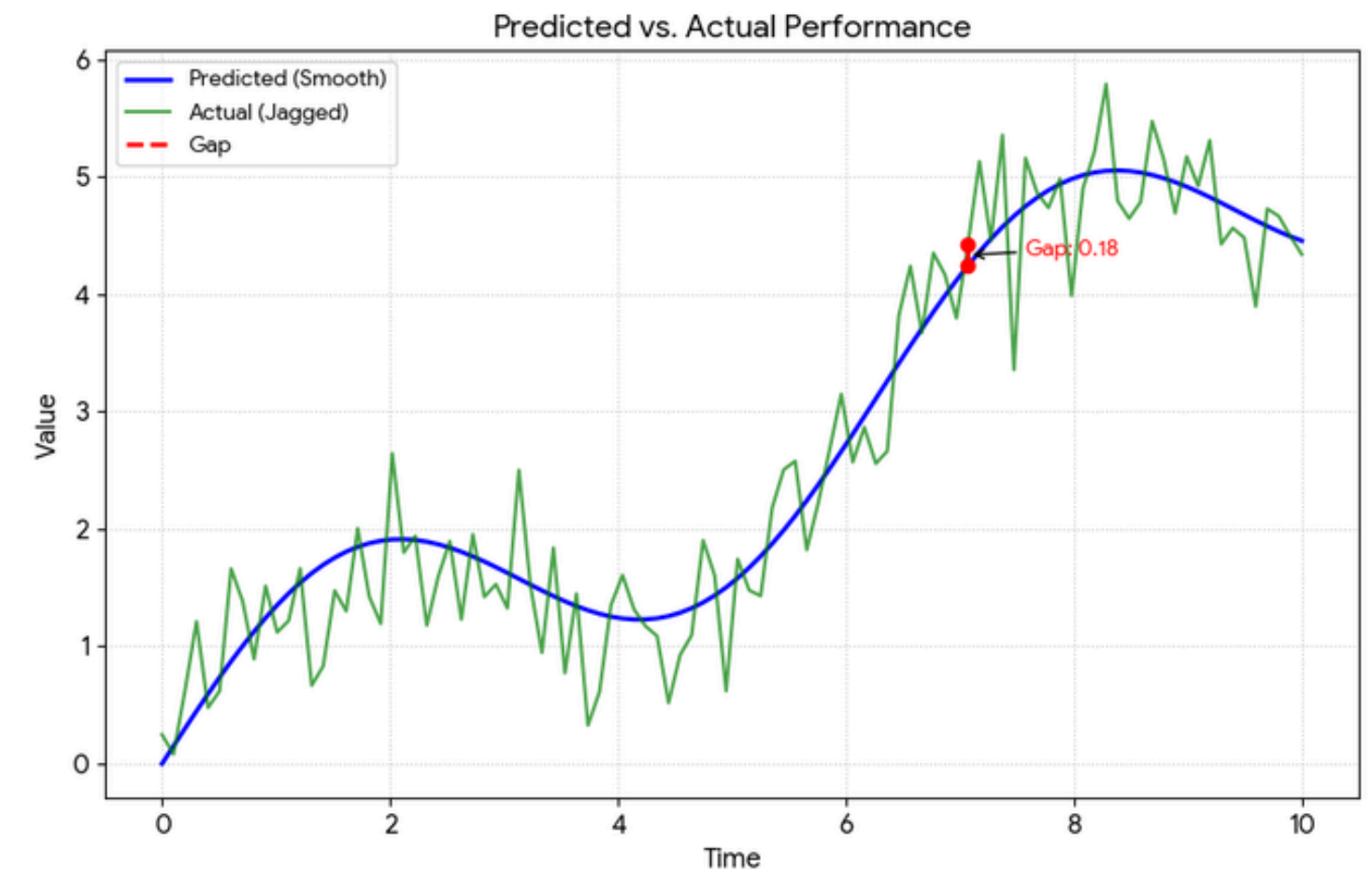
Self / Past Values

Autoregression



Shock

- **Prediction is rarely perfect.**
- Imagine you **predicted** the temperature would be **30°C**, but it was **actually 35°C**.
- That **+5°C difference** is the **Error**, also called the **Residual**, **Innovation**, or **Shock**.
- It represents new information or **unexpected events** that our **past-value model** couldn't see.



If a "**shock**" happened **yesterday** (e.g., a sudden market crash or a **breaking news story**), does its **impact disappear instantly**?

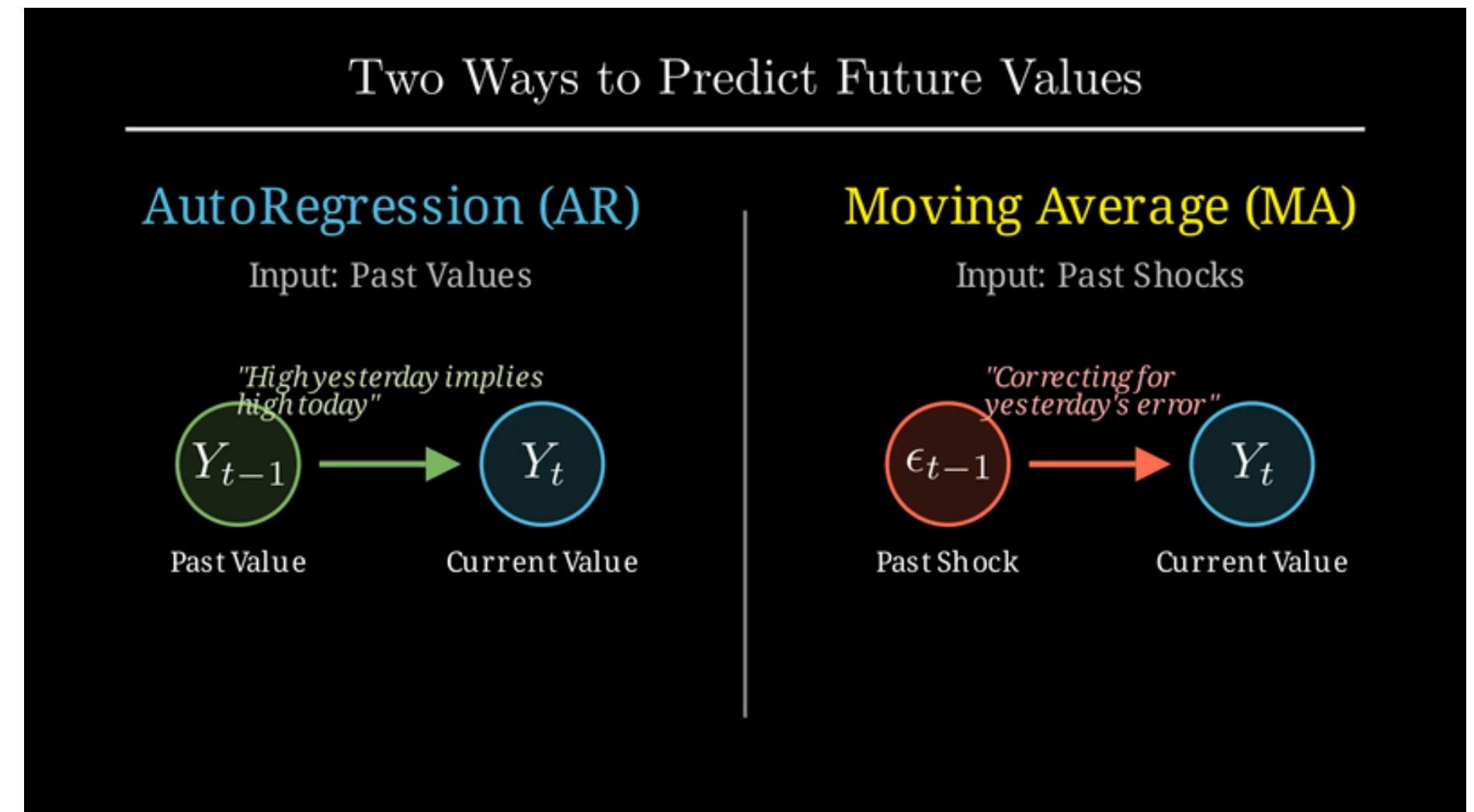
No. The "**ripple effect**" of that **shock** is often **felt today**.

Can we use these Past Errors to improve our Current Prediction?

Moving Average (MA)

- **Moving Average (MA)** model defines the current value of a time series as a **linear combination** of the **series mean** and **past white noise error terms** (shocks).

$$y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q}$$



Moving Average

- Coefficient (How much of yesterday's shock persists today?)

$$y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q}$$

Current Target Value

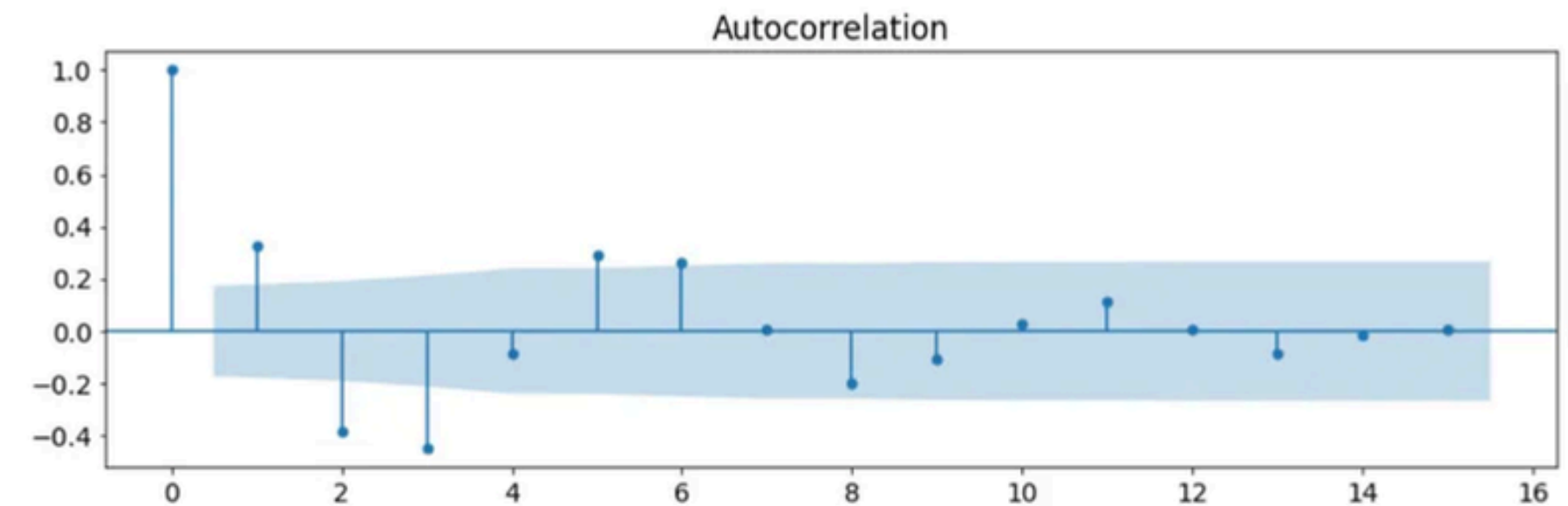
The average

Current Noise

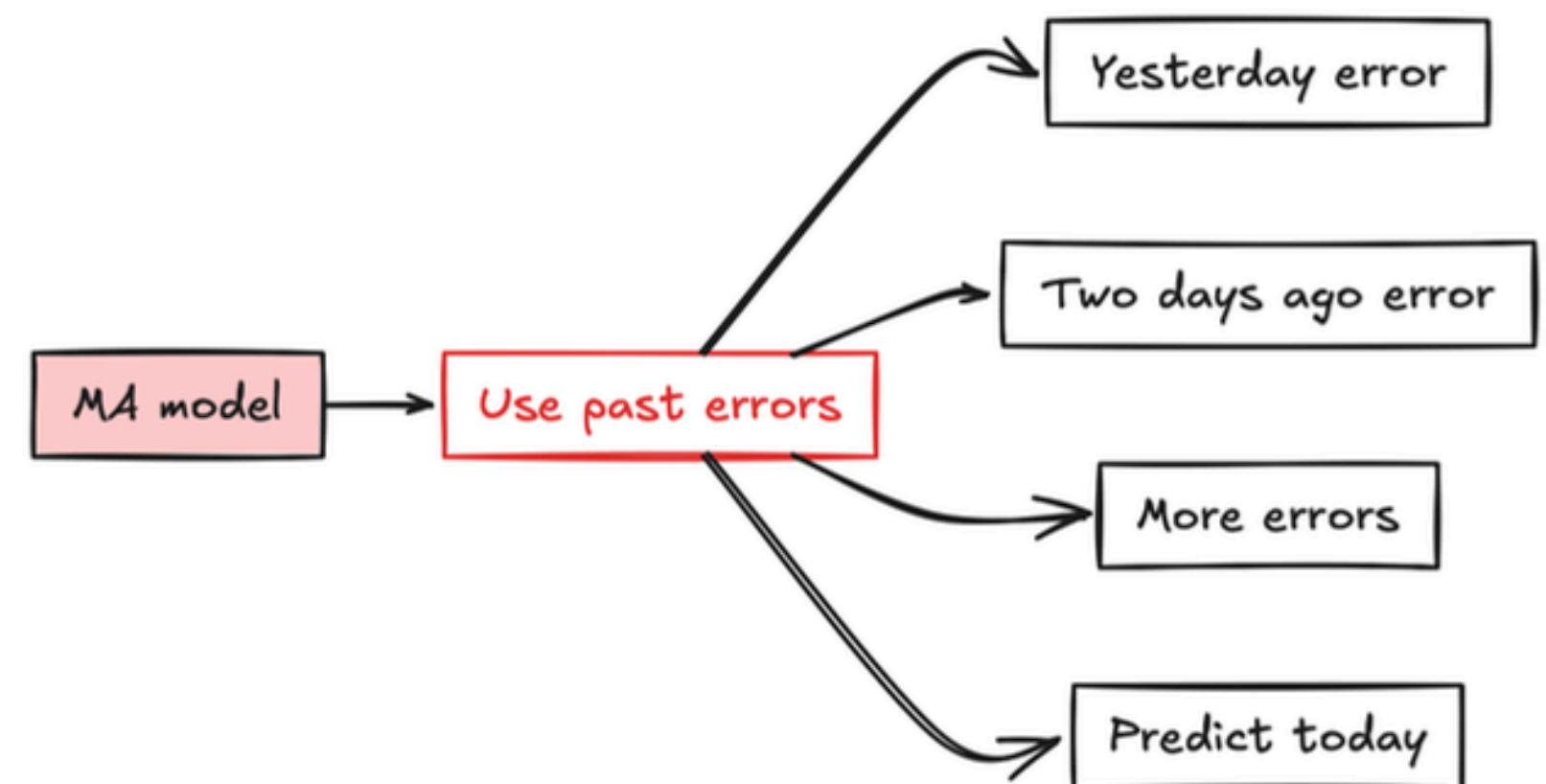
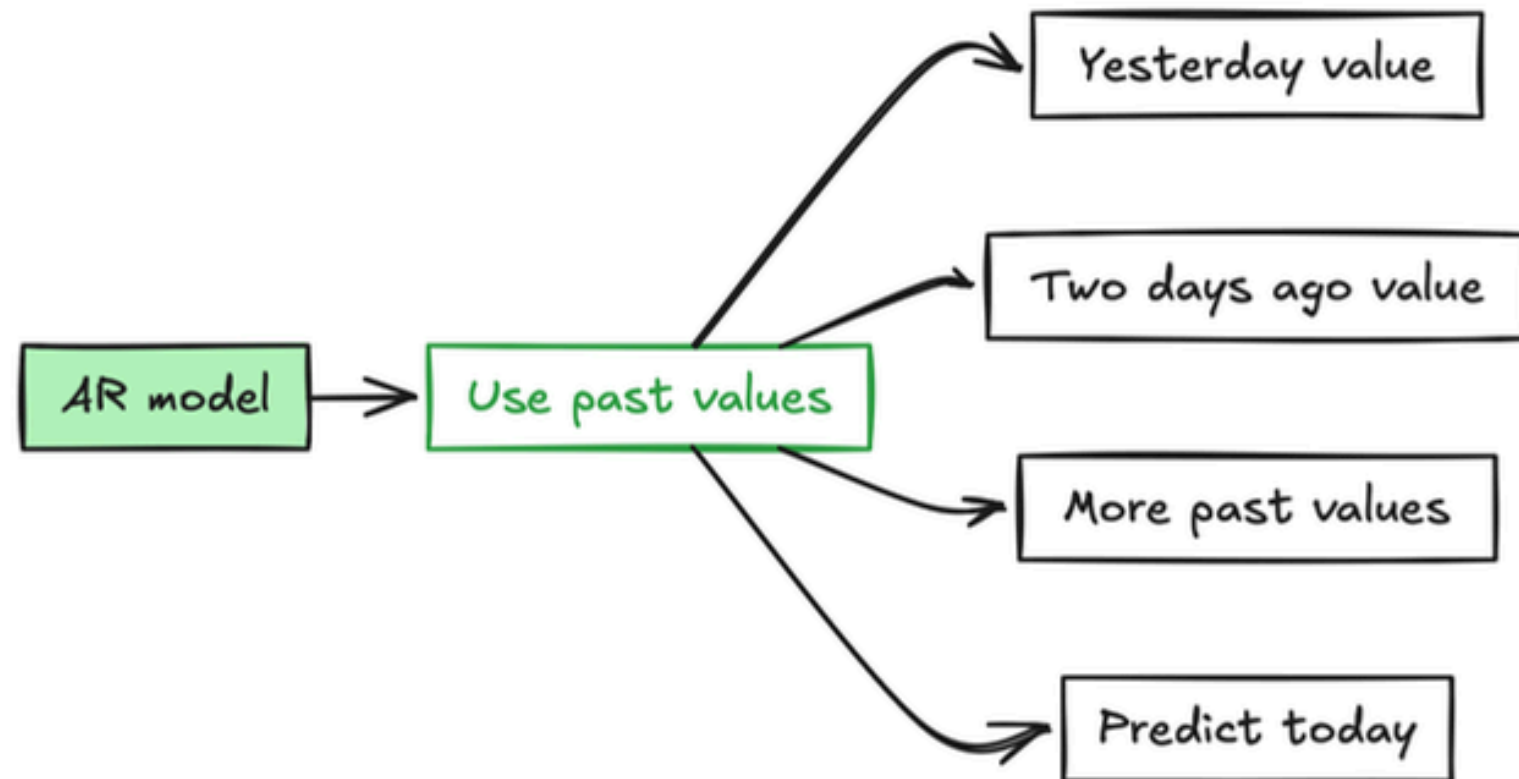
Past Shock (The error we made at t-1)

Moving Average

- To find which **past shocks matter**, we look at the **ACF Plot**.
- **Rule:** If the **ACF** shows a **significant spike** at **Lag 1** and then cuts off, it suggests an **MA model**.
- Key Difference:
 - **AR Model** – Check PACF
 - **MA Model** – Check ACF



AR vs MA



Both **AR and MA** assume the data is **Stationary**
(The **mean and variance stay constant**).

Most data has **Trends or Seasonality**. If the mean is constantly **changing**,
our **previous formulas break**.

Integration

- Integration is the process that removes non-stationarity.
- A time series is said to be **integrated** of order d —written as **$I(d)$** – if:
 - You must difference it d times
 - Before it **becomes stationary** (constant mean, variance, autocovariance)
- EXAMPLE:
 - A series that becomes stationary after one differencing is $I(1)$.
 - One that needs two differencing operations is $I(2)$.

Differencing

Day (t)	Sales X(t)	Lag1 X(t-1)	Diff1 = X(t) -X(t-1)	Diff2 (Difference of Diff 1)	Diff3 (Difference of Diff 2)
1	10	–	–	–	–
2	20	10	10	–	–
3	40	20	20	10	–
4	70	40	30	10	0
5	110	70	40	10	0

- **Order d = smallest differencing level that makes the differenced series stationary (stable).**
- Diff1 increases (10, 20, 30, 40)
- Diff2 becomes constant (10, 10, 10)
- Diff3 becomes 0 (0, 0)
- **Order is 2**

Differencing

t	X(t)	Lag2 = X(t-2)	Diff1 = X(t)- X(t-1)		Diff2 = Diff of Diff 1 X(t)-2X(t-
1	10				
2	25		15		
3	40	10	15	15	0
4	60	25	20	15	5
5	85	40	25	20	5

How to detect the needed level of integration?

- **Visual inspection**
 - Does the series have a trend? → Likely $I(1)$
- **Statistical tests**
 - ADF (Augmented Dickey–Fuller) test
 - KPSS test
 - PP test

- We have learned three separate concepts:
 - **AR (AutoRegressive)**: Predicting future values using past values.
 - **I (Integrated)**: Making data stationary by differencing.
 - **MA (Moving Average)**: Predicting future values using past errors (shocks).
- **ARIMA** stands for **AutoRegressive Integrated Moving Average**.

The ARIMA Equation

- **Differencing:** First, we calculate y'_t (the differenced data) to ensure stationarity.
- **Equation:** Then, we predict y'_t using both past values and past errors

$$y'_t = c + \underbrace{\phi_1 y'_{t-1} + \cdots + \phi_p y'_{t-p}}_{\text{AR terms}} + \underbrace{\theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}}_{\text{MA terms}} + \epsilon_t$$

Three Components of ARIMA

- An ARIMA model is defined by three parameters: $\text{ARIMA}(p, d, q)$.
 - **p (AR order)**: The number of lag observations included in the model (lag order).
 - **Question**: How many past days do I look at?
 - **d (Integration order)**: The number of times that the raw observations are differenced.
 - **Question**: How many times did I subtract to make the mean stable?
 - **q (MA order)**: The size of the moving average window (order of moving average).
 - **Question**: How many past error shocks do I include?

Please fill the feedback form.

Thank You